Neighborhood Label-Barycenter LMB Fusion for Distributed Multi-Target Tracking Under Unreliable Communication

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Abstract— Packet loss in a peer-to-peer labeled multi-Bernoulli (LMB) tracking network creates a component-correspondence failure: neighboring filters can assign different labels to the same target, or reuse one label for states that have drifted apart. Scalar arithmetic-average (AA) and Kullback–Leibler average (KLA) weights decide how much probability mass to trust, not which Bernoulli components are comparable before existence and spatial fusion. We introduce a graph-local neighborhood label-barycenter operator: each sensor chooses a median-cardinality medoid reference from its neighborhood, matches neighboring tracks to that reference by assignment, and replaces matched Gaussian posteriors by a first-two-moment barycenter. This construction separates label-set canonicalization from posterior averaging, enabling a reference-only ablation that tests whether label agreement alone explains the gain. Across 50 paired trials in a tiered packet-loss formation scenario, the operator reduces network optimal sub-pattern assignment (OSPA) disagreement by 81.59%, local expected OSPA (E-OSPA) by 17.15%, and local root-mean-square error (RMSE) by 6.35% relative to a tuned spatial-KLA AA baseline, while the reference-only ablation gives only 0.54% RMSE reduction. A held-out 50-trial replication preserves this full-versus-reference separation, with 6.64% RMSE reduction for the full operator versus 0.82% for reference-only label copying; fixed-parameter checks under harsher packet loss, ring topology, and partial field of view (FOV) preserve the same interpretation.

Index Terms— Arithmetic average fusion, distributed sensor networks, labeled multi-Bernoulli filter, multi-target tracking, random finite sets.

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This study uses simulated multi-target tracking data and no human-subject data.

Code and reproducibility artifacts are available at [repository DOI/URL].

I. INTRODUCTION

DISTRIBUTED multi-target tracking must estimate a time-varying target set while keeping local sensor estimates mutually usable under lossy communication. In networked surveillance and cooperative sensing systems, a local track is useful to a neighbor only when its identity and state are comparable with that neighbor's track. Labeled random finite sets (RFSs) and the labeled multi-Bernoulli (LMB) filter represent target birth, persistence, death, and track identity [1], [2]. In a peer-to-peer network, however, labels also encode local filtering history: after asymmetric measurements and packet drops, two sensors can assign different labels to the same target, or reuse one label for states that have drifted apart. The label set therefore becomes a fusion interface, not just bookkeeping.

Distributed RFS fusion is usually built around conservative geometric rules, such as generalized covariance intersection or the Kullback–Leibler average (KLA), or around arithmetic-average (AA) rules that preserve weak components and support heterogeneous labeled and unlabeled filters [3]–[6]. These rules allocate probability mass across neighbors. They do not, by themselves, decide which Bernoulli component in one local LMB posterior corresponds to which component in another. Once unmatched components enter the existence and spatial fusion consumers, better scalar weights can attenuate a bad contribution but cannot create the missing correspondence. The output can then keep a plausible cardinality while averaging states from different targets.

We target this residual correspondence failure directly. This paper asks a narrower question: once target-wise scalar weights are routed to the existence and spatial fusion consumers, can a graph-local projection construct the missing component map and average only matched posterior states? The comparison therefore uses a tuned AA-LMB baseline in which fusion weights are resolved at the object level before the existence and spatial consumers, so the proposed gain is not attributed to scalar-weight routing. The remaining design question is not another scalar-weight search: it is how to choose a neighborhood reference label set and how to replace posterior states after correspondence is established.

We propose a neighborhood label-barycenter operator for distributed LMB fusion. Each sensor chooses a median-cardinality medoid reference from its communication neighborhood, matches neighboring tracks to the reference labels using the Hungarian assignment method [7], and replaces each matched Gaussian group by its empirical first two moments. The operator is an output-space projection: repeating it propagates label-space agreement over the graph without reading a global label set or replacing the upstream Bernoulli existence update.

Contributions. The paper contributes four linked pieces: (i) a residual AA-LMB failure mode in which target-wise scalar weights are routed correctly but component labels still disagree; (ii) a graph-local medoid-assignment moment-barycenter operator that canonicalizes the neighborhood label set before averaging matched posterior states; (iii) structural conditions for cardinality robustness, stable assignment, reference-label invariance, and centralized moment-consensus limits; and (iv) paired 50-trial validation, a held-out 50-trial replication, fixed-parameter harsh-loss, topology-ring, and partial field-of-view (FOV) N50 checks, and a reference-only ablation that separates label copying from matched posterior barycentering.

II. RELATED WORK

Labeled RFS filters provide a Bayesian finite-set formulation for preserving target identity while estimating multi-object states [1]. The LMB filter gives a tractable approximation in which each Bernoulli component carries an existence probability and a labeled single-target density [2]. Multi-sensor and distributed versions of labeled RFS filters have been studied through generalized covariance-intersection fusion, robust distributed labeled-RFS fusion, consensus LMB updates, and resilient peer-to-peer LMB fusion [4], [5], [8], [9]. These methods address how sensors should share uncertain multi-object information, but practical labeled fusion still depends on whether components with different local labels are comparable.

KLA and logarithmic opinion-pool fusion are widely used because they avoid double counting under unknown correlations and connect naturally to consensus on probability densities [3]. This conservatism is valuable in distributed tracking, but geometric fusion can suppress weak components when local support is asymmetric. Arithmetic-average fusion follows a linear-pooling principle and has been analyzed statistically and information-theoretically for RFS densities [10]. Subsequent AA density-fusion work gives unified derivations for unlabeled and labeled RFS fusion, heterogeneous unlabeled/labeled fusion, and variational heterogeneous distributed fusion [6], [11], [12]. Fisher-information weighted AA fusion for multi-rate filters and weighted peer-to-peer RFS fusion further show that recent work is moving from one-shot density pooling toward communication- and filter-quality-aware AA designs [13], [14].

Heterogeneity and field-of-view mismatch create a separate difficulty. RFS fusion across heterogeneous sensors can be formulated without assuming identical local multi-object densities [15], and labeled RFS fusion with different fields of view has been studied explicitly [16]. Information-geometric labeled-RFS fusion offers another route for distributed heterogeneous state estimation [17]. These lines motivate robust fusion under nonidentical local views, but they do not by themselves remove the output-level failure in which two local LMB filters carry different labels for the same physical target after packet loss.

Our method is complementary to AA, KLA, and information-geometric weighting. The gap is not how to reweight an already matched density, but how to construct the correspondence map that makes local Bernoulli components comparable before output fusion. The contribution is therefore not another density-pooling rule: it is a targeted output-space projection, orthogonal to those weighting choices, that chooses a graph-local reference label set and computes matched moment barycenters. Finite-set fusion also raises consistency questions that are not present in single-target Gaussian fusion: pointwise consistency and global cardinality can move in different directions [18]. We therefore report both network disagreement and truth-referenced tracking error using optimal subpattern assignment (OSPA)-style metrics [19]. The method relies on assignment, for which the Hungarian method remains the standard polynomial-time solver [7], and on graph-local iteration, which is closely related to average-consensus ideas in sensor networks [20].

III. PROBLEM FORMULATION

Consider a network of S sensors. At time k, sensor s outputs an LMB estimate

$$X_s(k) = \{(\ell_{s,i}, r_{s,i}, \mu_{s,i}, \Sigma_{s,i})\}_{i=1}^{n_s(k)}, \tag{1}$$

where $\ell_{s,i}$ is a label, $r_{s,i} \in [0,1]$ is existence probability, and $(\mu_{s,i}, \Sigma_{s,i})$ are the first two moments of a Gaussian state approximation. Let $\mathcal{N}_s$ denote the sensor indices available to sensor s through the communication graph, including s itself.

The core difficulty is that the sets $\{\ell_{s,i}\}$ need not agree across sensors. A local filter may keep a label after a missed packet while a neighboring filter deletes it, or two sensors may assign different labels to the same physical target. Direct AA or KLA fusion of unmatched Bernoulli components then mixes component identities and states. We seek an estimate-level operator

$$\mathcal{B}_s : \{X_j(k) : j \in \mathcal{N}_s\} \mapsto Y_s(k) \tag{2}$$

that is local to the graph, preserves an interpretable reference label set, and reduces both pairwise output disagreement and truth-referenced tracking error.

The problem is not that scalar AA/KLA weights are unimportant. They remain the mechanism for allocating trust across incoming densities once the components being compared are known. The missing object is a correspondence map from local components to a common reference. The operator below estimates that map from the output geometry before applying a moment projection. In the reported implementation, this projection is applied to active output tracks after the upstream AA existence update and thresholding. It therefore rewrites the label set and Gaussian moments of the surviving tracks; it does not replace the Bernoulli existence consumer.

IV. NEIGHBORHOOD LABEL-BARYCENTER FUSION

A. Reference Selection

For each sensor s and time k, the candidate neighborhood is $\mathcal{N}_s$. Let $n_j(k)$ be the number of tracks output by neighbor j. The reference cardinality is chosen around the sample median,

$$m_s(k) = \text{median}\{n_j(k) : j \in \mathcal{N}_s\}. \tag{3}$$

The candidate set is

$$\mathcal{C}_s(k) = \arg\min_{j\in\mathcal{N}_s} |n_j(k) - m_s(k)|. \tag{4}$$

Among these candidates, the reference is the medoid under an OSPA-like set distance d_c with cutoff c:

$$\rho_s(k) = \arg\min_{j\in\mathcal{C}_s(k)} \frac{1}{|\mathcal{N}_s|} \sum_{q\in\mathcal{N}_s} d_c(X_j(k), X_q(k)). \tag{5}$$

The labels of $X_{\rho_s(k)}(k)$ form the local reference label set. In the implementation, d_c is computed from position states by Hungarian matching, squared matched distances, and a cutoff penalty c^2 for unmatched tracks, normalized by the larger cardinality. The cutoff is the configured consensus cutoff when available and otherwise follows the OSPA cutoff used by the tracking evaluation.

B. Assignment-Based Label Canonicalization

Let the reference have states $\{\mu_{\rho,a}\}_{a=1}^m$. For each neighbor j, we construct a cost matrix

$$C_{ba} = \|p(\mu_{j,b}) - p(\mu_{\rho,a})\|_2, \tag{6}$$

where $p(\cdot)$ extracts position coordinates. The Hungarian method returns the minimum-cost assignment between neighbor states and reference states [7]. This assignment defines the matched group $\mathcal{G}_{s,a}(k)$ for each reference label a.

C. Matched Moment Barycenter

For a matched group $\mathcal{G}_{s,a}(k) = \{(\mu_i, \Sigma_i)\}_{i=1}^M$, the output state is

$$\bar{\mu}_{s,a} = \frac{1}{M}\sum_{i=1}^M \mu_i, \tag{7}$$

$$\bar{\Sigma}_{s,a} = \frac{1}{M}\sum_{i=1}^M \left[\Sigma_i + (\mu_i - \bar{\mu}_{s,a})(\mu_i - \bar{\mu}_{s,a})^T\right]. \tag{8}$$

The output $Y_s(k)$ keeps the reference active labels and replaces their Gaussian moments by $(\bar{\mu}_{s,a}, \bar{\Sigma}_{s,a})$.

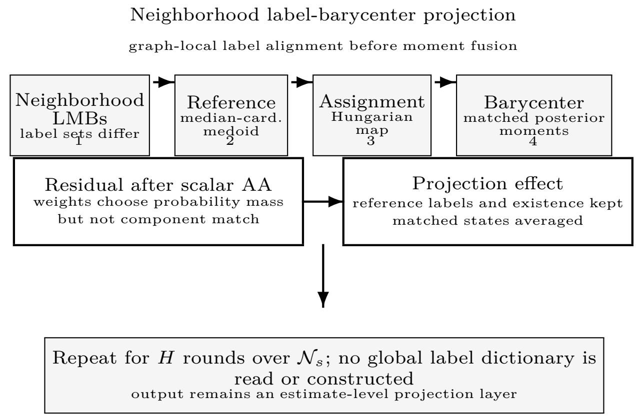

Fig. 1. Core mechanism of the proposed operator. Fusion weights specify how much probability mass a neighbor contributes, but they do not define component correspondence across local LMB posteriors. The operator therefore chooses a graph-local reference, matches neighbor tracks to that reference, and replaces matched states by posterior moment barycenters.

The projection does not estimate new Bernoulli existence probabilities: it is applied after upstream AA existence fusion and thresholding, passes the surviving active-track existence scores through with the selected reference labels, and rewrites only the label and moment fields. The implementation repeats this neighborhood operator for $H = 3$ rounds in the reported experiments.

D. Graph Locality and Computational Cost

The operator is graph-local at every round: sensor s only reads estimates indexed by $\mathcal{N}_s$, and no global label dictionary is constructed. The communicated object is the active-track tuple (ℓ, r, μ, Σ) already exchanged for output-level AA; the projection only adds local reference selection, assignment, and moment accumulation after reception. An H-round distributed implementation therefore requires H neighborhood output-exchange rounds and no centralized label service, while existence scores continue to come from upstream AA. If $d_s = |\mathcal{N}_s|$ and $n_{\max}$ upper bounds the number of tracks in the neighborhood, the dominant cost is assignment. The medoid stage evaluates set distances among neighborhood estimates, while the matching stage solves at most d_s rectangular assignments to the chosen reference. With a cubic Hungarian implementation, the per-round cost is bounded by $O(d_s^2 n_{\max}^3 + d_s n_{\max}^3)$ in the worst case, plus linear moment accumulation over matched tracks. This cost is acceptable in the reported formation setting, but it explains why the method is slower than scalar-weight AA and motivates the runtime study reported with the experimental results.

Input: neighborhood estimates, cutoff c, rounds H.
1 Reference: choose the median-cardinality OSPA medoid $\rho_s(k)$.
2 Match: assign neighbor tracks to reference labels by Hungarian matching.
3 Project: compute moment barycenters for matched states.
4 Existence: pass through upstream active-track existence scores; rewrite labels/moments only.
5 Iterate: overwrite active outputs locally; no global label dictionary is read.

Fig. 2. Implementation outline used in the validation. The operator is an estimate-level projection layer: it consumes only graph-neighborhood outputs, does not read a global label set, and does not re-estimate Bernoulli existence probabilities.

V. STRUCTURAL PROPERTIES

The following statements separate what the projection guarantees algebraically from what remains a tracking problem. They are stated under fixed output estimates at one time step, after the upstream AA existence consumer has selected active tracks.

Assumption 1 (Analysis scope):
Unless a proposition states otherwise, the structural claims below analyze a fixed neighborhood at one time step. They treat reference selection and assignment as operating on active tracks, hold births and deaths outside the projection layer, and do not assume that close target crossings or cardinality mismatches are resolved by the barycenter itself.

Proposition 1 (Weighting is not matching):
Consider two sensors and two one-dimensional targets with distinct means $x_1 \neq x_2$. Sensor 1 labels them (a, x_1) and (b, x_2), while sensor 2 labels them (a, x_2) and (b, x_1). A label-keyed scalar fusion consumer that fuses equal labels with weights $\alpha_a, \alpha_b \in [0, 1]$ produces means

$$\hat{x}_a = \alpha_a x_1 + (1-\alpha_a)x_2, \quad \hat{x}_b = \alpha_b x_2 + (1-\alpha_b)x_1. \quad (9)$$

If both sensors have positive weight for a component, the fused mean lies between the two physical targets rather than selecting the missing permutation. Choosing an endpoint weight avoids mixing only by discarding one sensor for that component. Thus scalar weights can suppress or emphasize an incoming component, but they do not by themselves infer the correspondence $a \leftrightarrow b$ created by the label swap.

Proof:
The displayed equations are the scalar-weighted means used by a label-keyed consumer in this two-component example. For $0 < \alpha_a < 1$, $\hat{x}_a$ is a strict convex combination of x_1 and x_2; the same holds for $\hat{x}_b$ when $0 < \alpha_b < 1$. The only way to recover an endpoint for a component is to set the corresponding weight to 0 or 1, which selects one sensor's label assignment rather than estimating the cross-sensor permutation. ■

Proposition 2 (Median-cardinality robustness):
If more than half of the sensors in $\mathcal{N}_s$ output the same cardinality $n^\star$ at time k, then the reference selected by the median-cardinality stage also has cardinality $n^\star$.

Proof:
With a strict majority at $n^\star$, the sample median of neighborhood cardinalities is $n^\star$. The minimum value of $|n_j(k) - m_s(k)|$ is therefore zero, and it is attained only by sensors with $n_j(k) = n^\star$. The subsequent medoid step is restricted to this candidate set. ■

Proposition 3 (Sufficient assignment stability):
Let the reference positions be $\{z_a\}_{a=1}^m$ with separation

$$\Delta = \min_{a \neq b} \|z_a - z_b\|_2. \quad (10)$$

Suppose a neighbor has the same cardinality and contains one state y_a for each reference target such that $\|y_a - z_a\|_2 \leq \epsilon$ after the physical correspondence is indexed by a. If $\epsilon < \Delta/(m+1)$, then the minimum-sum Euclidean assignment to the reference is the physical correspondence.

Proof:
The physical correspondence has assignment cost at most $m\epsilon$. Any nonphysical permutation assigns at least one state y_a to a different reference z_b, $b \neq a$. For that pair,

$$\|y_a - z_b\|_2 \geq \|z_a - z_b\|_2 - \|y_a - z_a\|_2 \geq \Delta - \epsilon. \quad (11)$$

All other assignment costs are nonnegative, so the nonphysical assignment costs at least $\Delta - \epsilon$. The condition $\epsilon < \Delta/(m + 1)$ gives $\Delta - \epsilon > m\epsilon$, hence every nonphysical assignment has larger cost. ■

Proposition 4 (Moment projection):
For a matched group $\mathcal{G} = \{(\mu_i, \Sigma_i)\}_{i=1}^M$, the barycenter mean minimizes $\sum_i \|z - \mu_i\|_2^2$, and, for any representative z,

$$\sum_i \|\mu_i - z\|_2^2 = \sum_i \|\mu_i - \bar{\mu}\|_2^2 + M\|z - \bar{\mu}\|_2^2. \quad (12)$$

Thus the common replacement $\bar{\mu}$ is the least-squares representative of the matched group and removes within-group mean disagreement after correspondence is fixed. The pair $(\bar{\mu}, \bar{\Sigma})$ gives the first two moments of the equally weighted Gaussian mixture $M^{-1} \sum_i \mathcal{N}(\mu_i, \Sigma_i)$.

Proof:
The identity follows by writing $\mu_i - z = (\mu_i - \bar{\mu}) + (\bar{\mu} - z)$ and using $\sum_i (\mu_i - \bar{\mu}) = 0$. It also shows that the least-squares objective is minimized uniquely at $z = \bar{\mu} = M^{-1} \sum_i \mu_i$. The covariance identity follows from total covariance:

$$\mathrm{Cov}(x) = \mathbb{E}_i[\mathrm{Cov}(x|i)] + \mathrm{Cov}_i(\mathbb{E}[x|i]). \quad (13)$$

■

Proposition 5 (Stable-matching consensus limit):
For one matched reference label, suppose the correspondence is fixed and correct over a connected graph, and let

$$q_s^h = \left[(\mu_s^h)^T, \mathrm{vec}(M_s^h)^T\right]^T, \quad M_s^h = \Sigma_s^h + \mu_s^h(\mu_s^h)^T. \tag{14}$$

If $q_s^{h+1} = \sum_j W_{sj} q_j^h$ with a primitive doubly stochastic matrix W respecting the graph, then all q_s^h converge to $S^{-1}\sum_j q_j^0$. The limiting mean and second moment therefore equal the centralized equal-weight moment barycenter.

Proof:
Stack the q_s^h vectors. Since $W^h \to S^{-1}\mathbf{1}\mathbf{1}^T$, every node converges to the average initial moment vector. Recovering Σ from $M - \mu\mu^T$ gives the covariance. ■

Proposition 6 (Upstream AA existence convexity):
If an AA existence update uses nonnegative weights that sum to one, then the fused existence probability remains in the convex hull of the incoming existence probabilities.

Proof:
The fused value is a convex combination of values in $[0, 1]$. ■

Proposition 7 (Reference-label invariance):
For every sensor s, the label set of $Y_s(k)$ is a subset of the label set produced by one sensor in $\mathcal{N}_s$. Moreover, if every estimate in $\mathcal{N}_s$ has the same labels and identical matched Gaussian moments, then $\mathcal{B}_s$ returns that estimate unchanged.

Proof:
The operator defines output labels only from the selected reference $X_{\rho_s(k)}(k)$, so it never creates a label outside a neighborhood estimate. If all neighborhood estimates share the same labels and moments, every admissible reference has the same label set, each assignment is the identity up to ties among identical states, and each matched group contains identical moments. The barycenter equations therefore return the same mean and covariance. ■

These properties are structural, not a blanket tracking-optimality claim. The assignment-stability condition is sufficient rather than necessary, and it excludes cardinality mismatch, births, deaths, and close target crossings. The consensus-limit result concerns the matched moment coordinates and is conditional on stable matching and stochastic graph weights; the reported implementation uses three finite rounds and recomputes references and assignments. The existence convexity statement belongs to the upstream AA existence consumer, which the projection layer leaves intact. The propositions instead motivate two testable predictions: reference-only canonicalization should reduce label-space disagreement when correspondence is the dominant error, while matched moment barycenters should add truth-referenced localization gains when state disagreement remains after labels are aligned. The empirical section therefore holds scalar AA routing fixed and compares the tuned baseline, the reference-only label-copying ablation, and the full projection.

VI. EXPERIMENTAL SETUP

The evaluation uses an eight-sensor formation-tracking scenario with tiered fixed packet-loss probabilities. Each 50-trial paired run uses the same seeds and the same per-trial packet-loss assignments across methods. The loss levels are $\{0, 0.1, 0.2, 0.5\}$, with one sensor at 0, four sensors at 0.1, one sensor at 0.2, and two sensors at 0.5 in each trial. The LMB filters use AA parallel update mode, Metropolis fusion weights, maximum three Gaussian-mixture components, 150 loopy-belief-propagation iterations, and an existence threshold of 0.18.

We compare three arms chosen to isolate the mechanism rather than to search for scenario-specific hyperparameters. The baseline is a tuned spatial-KLA AA implementation with target-wise weight handling, link-quality weighting, existence confidence, and structure-aware spatial decoupling; it tests whether label-space errors remain after scalar fusion weights are routed to the correct consumers. The proposed arm keeps that upstream existence pathway and enables neighborhood label-barycenter projection over communication neighborhoods with $H = 3$ fixed iterations; it tests the full label-canonicalization and matched-barycenter hypothesis on the active output tracks. The ablation arm uses the same reference selection and label canonicalization but copies the medoid reference states without matched-state barycenters; it tests whether label-set agreement alone explains the gain. All arms keep their parameters fixed across the 50 paired trials and see the same measurements and packet-loss realization in each trial.

Network disagreement is measured by pairwise output OSPA, localization disagreement, and cardinality dispersion across sensors. Truth-referenced tracking quality is measured by local expected OSPA (E-OSPA), root-mean-square error (RMSE), and cardinality error averaged across sensors. OSPA is used because it consistently combines localization and cardinality errors for multi-object filters [19]. We report mean values, 95% confidence intervals over trials, paired reductions relative to the tuned baseline, wins out of 50 paired trials, and sign-test p values.

The experiment suite is organized around mechanism claims rather than parameter search. The primary 50-trial paired run tests whether a correspondence projection improves a tuned AA baseline when scalar weight routing is already handled.

TABLE I
Mechanism-isolation protocol for the AA comparison.

Arm	Enabled mechanism	Hypothesis tested
Tuned spatial-KLA AA	Target-wise scalar weights and spatial-KLA AA	Correct weight routing alone does not solve component correspondence.
Reference-only	Neighborhood reference selection and label copying	A common label set can reduce disagreement but should not explain the main spatial gain.
Label-barycenter	Reference selection, assignment, and matched moment barycenters	Matched posterior barycenters improve both agreement and truth-referenced tracking.

The reference-only ablation tests whether choosing a common neighborhood label set is sufficient, or whether matched posterior moments are needed for localization gains. The held-out and boundary runs are fixed-parameter checks; they vary seeds, packet-loss severity, topology, or sensing geometry only after the method settings have been fixed.

With base seed 1, the 50 trial seeds are 2 through 51. Each trial samples one eight-sensor packet-loss vector from the tiered profile and reuses it for all arms. Intervals are mean ± 1.96 standard errors over trial-level means; paired reductions use baseline-minus-candidate deltas. Wins and two-sided sign-test values are computed on nonzero paired deltas.

The operator parameters are fixed before the reported N50 evaluation. In particular, the neighborhood version uses $H = 3$ graph-local rounds, the same existence threshold as the tuned AA baseline, and no per-scenario search over projection cutoffs, barycenter weights, or trial-specific label rules. Held-out base-seed runs are reserved for robustness checks and are not used to choose these settings.

Runtime is measured as wall-clock filter time from one GNU Octave 11.1.0 execution on an Apple M4 workstation with 16 GB memory. Absolute seconds characterize this MATLAB-compatible prototype; the relative column is the intended paired cost comparison.

The numerical tables and Fig. 3 are generated from tracked validation reports rather than edited by hand. The reproducibility package records the source report path, source SHA-256 digest, generated machine-readable summaries, and a generated ledger that separates paired AA evidence, held-out robustness evidence, harsh-loss stress evidence, topology/FOV boundary checks, and contextual GA rows. A verifier recomputes network-disagreement statistics from per-trial report rows, runtime statistics from trial logs, and local tracking metrics from per-trial local rows before the PDF is built.

VII. RESULTS

Table II shows that the proposed operator improves the communication-facing and truth-facing metrics simultaneously. For network OSPA, the 95% confidence intervals are $[1.667, 1.699]$ for the tuned baseline, $[0.300, 0.319]$ for the full method, and $[1.014, 1.046]$ for the reference-only ablation. For local E-OSPA, the corresponding intervals are $[2.013, 2.047]$, $[1.664, 1.699]$, and $[1.893, 1.929]$. The separation is therefore visible in both output agreement and truth-referenced tracking before paired tests are considered.

Table III places the proposed AA projection beside two GA reference rows generated with the same base seed, trial seeds, and tiered packet-loss profile. Because those rows come from the GA validation path, they are used only as contextual baselines rather than as paired significance-test inputs. In this setting, the neighborhood label-barycenter row is lower than both GA reference rows on all six reported disagreement and tracking metrics, which supports the interpretation that label canonicalization and matched posterior barycenters address a failure mode not removed by the existing GA weighting variants alone.

Table IV and Fig. 3 separate label-set effects from state-barycenter effects. The reference-only ablation reduces network disagreement because sensors in a neighborhood move toward a common label set, but its RMSE reduction is only 0.54%, with a confidence interval crossing zero, 27 wins out of 50 paired trials, and no sign-test support. The full operator reaches 6.35% RMSE reduction, with 48 wins out of 50 and $p < 10^{-3}$. Cardinality dispersion and cardinality error reductions are identical for the reference-only and full arms because both use the same reference-label projection; these metrics therefore isolate the label-set effect rather than the moment-barycenter effect. The full method also improves local E-OSPA in all 50 paired trials. These results support the claim that matched posterior barycenters, not label copying alone, drive the spatial tracking gain.

The held-out base-seed run repeats the mechanism test without changing the method parameters. On this run, the full operator reduces network OSPA by 81.85%, local E-OSPA by 17.15%, and local RMSE by 6.64% relative to tuned spatial-KLA AA. The full RMSE reduction has 48/50 wins with sign-test $p < 10^{-3}$, whereas the reference-only RMSE reduction is 0.82% with 28/50 wins and sign-test $p = 0.480$. The held-out evidence therefore preserves the same separation between label-set copying and matched posterior barycentering.

Runtime also increases: full/reference-only cost 1.644/1.534 times the tuned baseline, with the full

TABLE II

Fifty-trial paired validation under tiered packet loss. Lower values are better, and bold entries mark column-wise best values including ties. Values are trial means. The proposed operator improves both network agreement and local tracking quality; the reference-only ablation isolates the part of the gain that comes from label-set canonicalization alone.

Method	Net OSPA	Loc. disag.	Card. disp.	E-OSPA	RMSE	CardErr
Tuned spatial-KLA AA	1.683	1.473	0.035	2.030	3.683	0.088
Neighborhood reference-only	1.030	0.892	0.021	1.911	3.663	0.077
Neighborhood label-barycenter	0.310	0.216	0.021	1.681	3.449	0.077

TABLE III

Contextual reference rows from the same 50 trial seeds and tiered packet-loss profile. Lower values are better, and bold entries mark column-wise best values. The GA rows are generated by the tracked GA validation path and are not used in the paired AA sign tests.

Method	Net OSPA	Loc. disag.	Card. disp.	E-OSPA	RMSE	CardErr
GA +structure-aware KLA	1.761	1.463	0.084	2.213	4.228	0.204
GA +FID-FIA refinement	1.848	1.958	0.081	2.180	4.695	0.151
Neighborhood label-barycenter	0.310	0.216	0.021	1.681	3.449	0.077

TABLE IV

Paired reductions relative to the tuned spatial-KLA AA baseline. The bracketed interval is the 95% confidence interval of the absolute paired reduction, and p is the two-sided sign-test value on nonzero paired deltas.

Metric	Full reduction	Full wins	Full p	Ref.-only reduction	Ref.-only wins	Ref. p
Network OSPA	81.59% [1.359, 1.386]	50/50	$< 10^{-3}$	38.78% [0.639, 0.666]	50/50	$< 10^{-3}$
Loc. disagreement	85.31% [1.238, 1.275]	50/50	$< 10^{-3}$	39.46% [0.566, 0.596]	50/50	$< 10^{-3}$
Card. dispersion	40.03% [0.012, 0.016]	50/50	$< 10^{-3}$	40.03% [0.012, 0.016]	50/50	$< 10^{-3}$
E-OSPA	17.15% [0.342, 0.355]	50/50	$< 10^{-3}$	5.83% [0.112, 0.125]	50/50	$< 10^{-3}$
RMSE	6.35% [0.201, 0.267]	48/50	$< 10^{-3}$	0.54% [-0.017, 0.056]	27/50	0.672
Card. error	12.27% [0.009, 0.013]	46/50	$< 10^{-3}$	12.27% [0.009, 0.013]	46/50	$< 10^{-3}$

TABLE V

Held-out base-seed N50 robustness check. Reductions are relative to tuned spatial-KLA AA; bracketed intervals are 95% confidence intervals for the absolute paired reduction. The run uses base seed 11 and the same fixed three-arm protocol as the main validation.

Metric	Full red. [CI]	Full wins/p	Ref. red. [CI]	Ref. wins/p
OSPA	81.85% [1.367, 1.394]	50/50, $< 10^{-3}$	39.15% [0.647, 0.674]	50/50, $< 10^{-3}$
E-OSPA	17.15% [0.342, 0.357]	50/50, $< 10^{-3}$	5.88% [0.114, 0.126]	50/50, $< 10^{-3}$
RMSE	6.64% [0.210, 0.281]	48/50, $< 10^{-3}$	0.82% [-0.008, 0.069]	28/50, 0.480

method increasing from 42.487 s to 69.729 s, so assignment and output rewriting dominate moment computation.

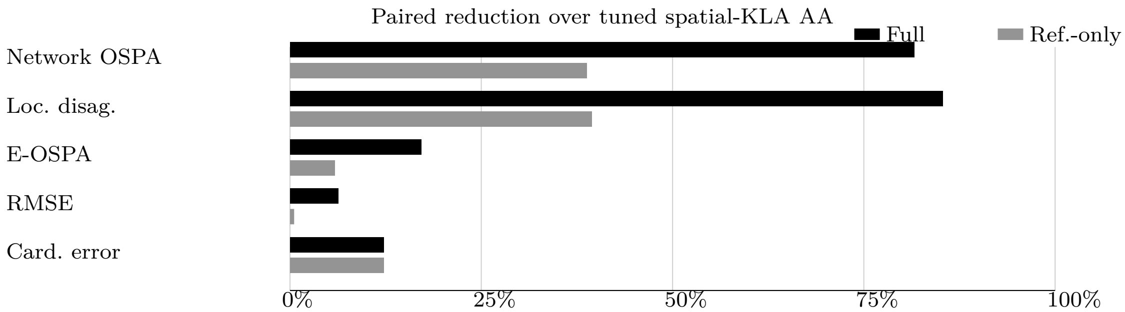

Fig. 3. Paired percentage reductions over the tuned spatial-KLA AA baseline. Black bars show the full label-barycenter operator, and gray bars show the reference-only ablation. The reference-only ablation captures the effect of choosing and copying a consistent local reference label set. The full method gives substantially larger localization and RMSE gains because it also averages matched posterior states.

VIII. DISCUSSION AND LIMITATIONS

Taken together, the paired and boundary evidence support a specific interpretation. When target-wise AA weight routing is held fixed, reference-only projection reduces network disagreement and captures the cardinality effect, but it leaves the main RMSE separation unsupported. Adding matched moment barycenters creates the spatial tracking separation across the primary, held-out, harsh-loss, topology-ring, and partial-FOV checks. The method should therefore be read as a correspondence-and-projection layer complementary to AA/KLA weighting, not as a replacement for density pooling.

This interpretation has boundaries. The operator remains an estimate-level projection layer. It is graph-local because each sensor uses only its communication neighborhood, but it is not a full recursive message update inside the LMB filtering loop. A recursive implementation would also need lifecycle guards for birth, death, and intermittently matched labels, because unstable matches can turn a useful barycenter into an erroneous track merge.

The main failure mode is assignment ambiguity rather than packet loss alone. When targets are closely spaced, cardinalities lack a neighborhood majority, or one local posterior is biased, the medoid can select a plausible but wrong correspondence. A recursive deployment should therefore gate projection by assignment-margin, covariance, track-age, or reliability evidence, and fall back to reference-only or upstream AA output when correspondence is not credible. We omit such guards here to isolate the label-map and moment-barycenter mechanisms.

The method uses Euclidean position matching and equally weighted moment barycenters. These choices keep the operator interpretable and make the reference-only ablation clean, but they do not use covariance quality, label age, or link reliability inside the barycenter itself. A covariance-aware or reliability-weighted variant is a natural extension when local uncertainty differs strongly across sensors. The main validation is concentrated on a tiered packet-loss formation scenario. As a packet-loss severity check, the same fixed parameters were evaluated under $p_{\mathrm{drop}} \in [0.2, 0.35, 0.5, 0.7]$ with sensor counts [1, 3, 2, 2]. The full operator keeps the mechanism separation in this harsher setting, reducing network OSPA by 79.57%, local E-OSPA by 17.06%, and RMSE by 7.98%, while the reference-only arm reduces RMSE by 2.47%. Additional fixed-parameter N50 scenario-family checks probe topology and sensing-geometry changes. In the topology-ring case (base seed 31, topology ring, FOV 60 deg), the full operator reduces network OSPA/local E-OSPA/RMSE by 47.50%/21.53%/13.12%, compared with a 5.17% RMSE reduction for reference-only. In the partial-FOV case (base seed 41, topology 4plus4, FOV 35 deg), the full operator reduces network OSPA/local E-OSPA/RMSE by 43.25%/6.89%/6.50%, compared with a 2.13% RMSE reduction for reference-only. The stress check broadens packet-loss severity, the topology-ring check broadens sparse-topology coverage, and the partial-FOV check broadens partial-field-of-view sensing; together, the evidence still preserves the same formation-family assumptions and should not be read as a substitute for maneuvering-target or covariance-consistency studies.

IX. CONCLUSION

This paper reframes a persistent AA-LMB fusion error as a label-space alignment and matched-posterior barycenter problem. The proposed neighborhood label-barycenter operator chooses a robust local reference, canonically matches neighboring tracks to that reference, and computes first-two-moment barycenters for matched states. In 50 paired trials under tiered packet loss, the method improves both network agreement and local tracking metrics, while a reference-only ablation shows that the main spatial gain comes from posterior barycentering rather than label copying alone. The resulting graph-local projection can support future recursive LMB designs with explicit lifecycle and consistency safeguards.

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